Research Proposal: An AI and Engineering-Driven Empowerment Solution for Neuroscience Research

I. Applicant Background and Research Vision

My name is Xiangru Zhou, with over 10 years of experience in software algorithm development, specializing in computer vision, AI model deployment, and system optimization. I have filed for or obtained more than 10 invention patents. I hold a Bachelor's degree in Microelectronics from Xi'an University of Technology and am currently pursuing a Master's degree in Microelectronics Technology & Materials at The Hong Kong Polytechnic University. During my master's studies, under the guidance of my supervisor, I gained in-depth experience with molecular dynamics AI platforms such as LAMMPS, laying a foundation for interdisciplinary research.

For a long time, I have been focusing on the application potential of artificial intelligence and computer technology in biomedical engineering and neuroscience. I firmly believe that the in-depth integration of engineering technology and basic scientific research will drive breakthroughs in neuroscience. As a core discipline exploring brain functions and revealing the mechanisms of sensory information processing and behavioral regulation, neuroscience faces challenges such as complex data analysis and experimental process automation, which are highly aligned with my technical expertise. I aim to transition into the interdisciplinary field of biomedical engineering and neuroscience by leveraging my strengths in image processing, AI system development, and software-hardware integration. Through a virtuous cycle of "biomedical algorithm development - engineering implementation - research scenario application - algorithm iteration and optimization", I intend to provide customized technical support for neuroscience research and help address core scientific questions such as "how sensory information is represented in the brain and transformed into behavioral responses".

II. Core Research Framework

Guided by the common technical needs of neuroscience research, this proposal constructs an interdisciplinary research system of "biological theory support - molecular dynamics simulation - engineering technology empowerment - research value transformation". Through the deep integration of technology and scientific research, it aims to improve research efficiency and achieve breakthroughs in mechanism exploration. The specific cycle is as follows:

1. **Biomedical Algorithm Optimization:** Develop customized image processing and AI analysis algorithms based on the needs of imaging data, behavioral data, and other research data in neuroscience;
2. **Engineering Implementation:** Integrate algorithms into software-hardware systems to build universal research tools with high automation and usability;
3. **Research Scenario Application:** Deploy and apply the tools in various neuroscience research areas such as visual neural circuits and sensory-behavioral regulation, collecting experimental data and feedback;
4. **Algorithm Iteration and Upgrade:** Continuously optimize algorithm performance and system adaptability based on practical needs and data feedback from scenario applications, expanding the scope of application.

III. Research Content and Implementation Plan for Each Module

(I) Biological Theory Learning and Knowledge Reserve

To ensure the accurate alignment of engineering technology with neuroscience research, I will systematically supplement relevant theoretical knowledge to provide a scientific basis for technical research and development:

1. **Neurobiology:** Focus on learning the composition and functions of sensory neural circuits such as visual and auditory systems, mastering core principles including neural coding mechanisms, neural circuit activation rules, and the transformation of sensory information into motor commands;

2. **Cell Biology:** Conduct in-depth study of neuronal structures, synaptic transmission processes (e.g., vesicle fusion, neurotransmitter release and binding), clarifying the dynamic response characteristics of neurons under sensory stimulation, and providing theoretical support for the localization and quantification of neuronal activity in image analysis;

3. **Molecular Biology:** Grasp the functions of key biomolecules such as neurotransmitters and receptor molecules, understand the regulatory mechanisms between gene expression and neural activity, and lay a foundation for the integration of molecular dynamics simulation and neural function analysis.

The learning method will combine coursework, intensive reading of core literature, and communication with research teams. I will carefully study top-tier journal papers in the field of neuroscience and regularly participate in academic discussions of the research team to ensure that theoretical knowledge is closely linked to research practice.

(II) Molecular Dynamics Simulation Empowering Neural Mechanism Exploration

Building on the experience with LAMMPS and molecular dynamics AI platforms accumulated during my master's studies, I will conduct targeted simulation research combined with core directions of neuroscience:

1. Focus on key biomolecules related to neural circuit functions (e.g., neurotransmitters, ion channel proteins, receptor molecules), and reproduce their dynamic behaviors (such as diffusion, binding, and conformational changes) during neural signal transmission through molecular dynamics simulation;
2. Optimize simulation parameters using AI models (e.g., DeePMD) to improve the accuracy and efficiency of molecular activity simulation, revealing the correlation between biomolecular dynamic changes and neuronal activation, as well as neural circuit function regulation;
3. Cross-validate simulation results with neuroimaging data (e.g., two-photon

imaging, functional magnetic resonance imaging) to provide molecular-level evidence for analyzing the activation mechanisms of neural circuits under sensory stimulation, assisting in addressing core scientific questions in neuroscience.

(III) Engineering Technology Research and System Construction

Based on 10 years of engineering experience, focusing on the common technical needs of neuroscience research, develop universal software-hardware systems to realize the automation and intelligent upgrading of research processes, including six specific modules:

1. **Integrated Image Processing:** For imaging data commonly used in neuroscience research (e.g., two-photon imaging, fluorescence imaging, functional magnetic resonance imaging), develop an end-to-end automated processing workflow that integrates image acquisition, data transmission, AI inference, result display and feedback, and UI interaction. Eliminating the need for separate operation of each link on different devices, it realizes one-click processing from raw images to core results such as neuronal activity quantification and neural circuit connection analysis. It supports key functions including automatic segmentation of neuronal regions in imaging data, extraction of neural signal features, and localization of activated neuron populations.
2. **Automatic Image and Data Acquisition and Classification:** Based on deep learning classification models (e.g., MobileNet, ResNet) and preset rules, automatically name, classify, and store behavioral images, neuroimaging data, physiological signal data, etc., generated from experiments. Establish a standardized classification system based on stimulus types, behavioral response patterns, experimental groups, etc., to provide high-quality datasets for subsequent model training, data validation, and data augmentation.
3. **User-Friendly Visual Interface Development:** Develop a visual operation interface using QT framework, encapsulating complex image processing algorithms, AI model inference processes, and data analysis functions. Design an intuitive operation logic that supports parameter adjustment, visual display of

results (e.g., neuron activation heatmaps, neural signal time-series curves), and data export, lowering the threshold for researchers without a computer science background and enabling wide adaptation of technical tools.

4. **Adaptive AI System Deployment and Adjustment:** Develop AI model versions adapted to different computing resources for computing environments in various research scenarios (e.g., local PCs, high-performance computing clusters, edge computing devices). Deploy lightweight models on devices with limited computing power to meet real-time data processing needs; run high-precision models on high-performance clusters to support in-depth analysis of large-scale data, ensuring full coverage of algorithms in various research scenarios.

5. **AI Model Acceleration and Optimization:** Combine technologies such as CUDA GPU acceleration, TBB multi-threading, and AVX instruction set optimization to optimize model training and inference speed for core tasks such as neuroimaging data processing, neuronal activity classification, and molecular dynamics simulation data analysis. Improve the efficiency of offline data processing by more than 30% and control AI inference latency in real-time monitoring scenarios within 10ms to ensure the efficient progress of experimental research.

6. **Software-Hardware Integrated System Development:** Develop miniaturized, modular biological experiment auxiliary devices using 3D printing technology, embedded sensors, and edge computing equipment. Realize communication and interconnection with various neuroscience experimental equipment (e.g., imagers, behavioral recording systems, electrophysiological devices), integrating functions such as imaging data acquisition, real-time AI inference, experimental status monitoring, and automatic data recording and classification to build a closed-loop experimental research platform.

(IV) Research Scenario Application and Value Transformation

Apply the developed technical tools and systems to various neuroscience research and related preclinical studies to realize the landing of technical value:

1. **Empowerment of Neurobiological Research:** Develop specialized analysis modules for neuroimaging data (e.g., two-photon imaging, fMRI, EEG) to extract key information such as functional connection patterns of sensory nervous systems, activation time-series of neuron populations, and neural signal features, quantifying differences in neural circuit activity under different experimental conditions; combine AI algorithms with molecular dynamics simulation results to assist in analyzing physiological and pathological correlation features in neural signals, providing data support for revealing the working mechanisms of neural circuits.
2. **Preclinical Research Application:** Extend the optimized image and neural data processing tools to preclinical research on neurological diseases such as Alzheimer's disease, epilepsy, and Parkinson's disease. By identifying abnormal neural activity patterns in disease models and quantifying pathological-related neural circuit changes, provide data support for the screening of disease diagnostic markers, pathological mechanism research, and optimization of treatment strategies, promoting the transformation from basic research to clinical application.

IV. Expected Outcomes and Value

(1) **Technical Outcomes:** Develop 1 set of universal integrated software-hardware systems for neuroscience research, including expandable image processing algorithm libraries, visual operation interfaces, data management platforms, and other core components; publish 1-2 academic papers in the interdisciplinary field of engineering technology and neuroscience, and file 2-3 related technical invention patents.

(2) **Research Support Outcomes:** Provide efficient and user-friendly customized data processing tools for neuroscience research teams, improving experimental data processing efficiency by more than 30% and reducing human operation errors; through the combination of molecular dynamics simulation and AI analysis, provide

new technical perspectives and data support for neural circuit mechanism research, assisting in addressing core scientific questions.

(3) **Long-Term Value:** The constructed "algorithm - engineering - research - algorithm" cycle model can be extended to neuroscience research on various model organisms (e.g., mice, Drosophila, C. elegans), providing a paradigm for interdisciplinary cooperation; cultivate research capabilities integrating engineering technology and neuroscience, laying a foundation for future clinical transformation research and promoting the innovative application of AI in the diagnosis and treatment of neurological diseases.

V. Cooperation and Implementation Guarantee

(1) **Interdisciplinary Collaboration:** Proactively communicate with neuroscience research teams to clarify specific research needs and technical indicators, ensuring that technical research and development are highly aligned with research objectives; regularly participate in team academic meetings, report technical progress, collect feedback, and dynamically adjust research directions.

(2) **Technical Implementation Guarantee:** Rely on 10 years of engineering development experience to ensure the stability and practicality of system development; use university laboratory resources and high-performance computing platforms to complete model training and system testing; leverage industrial resources to promote the rapid iteration and optimization of software-hardware prototypes.

(3) **Knowledge Reserve Upgrade:** Formulate a systematic learning plan for neuroscience knowledge, completing the supplement of core theoretical knowledge within 1 year after enrollment; actively participate in academic conferences and training in the fields of neuroscience and biomedical engineering to broaden research horizons and deepen interdisciplinary understanding.

VI. Conclusion

I firmly believe that with solid engineering technical accumulation, interdisciplinary learning capabilities, and a persistent pursuit of the neuroscience field, I can bring new technical vitality to various neuroscience research teams, while realizing my own transformation into the interdisciplinary field of biomedical engineering and neuroscience, contributing to promoting breakthroughs in neuroscience research and the innovative application of AI in the biomedical field.